

Objective:

You will design and implement a **C++ application for Raspberry Pi** that demonstrates:

- Multiple inheritance (**diamond shape**)
- Polymorphism
- Stream abstraction
- Hardware interaction (7-segment display)

The system must support **output redirection** either to:

- The **terminal**
- A **7-segment display**

based on the constructed object, while input is always read from the terminal.

Functional Requirements

1. Input

- Always taken from the terminal.
- Must be a **single digit (0–9)**.
- Invalid input must be rejected or handled gracefully.

2. Output

- Must be written using a **generic output stream interface**.
- Output destination is determined **at runtime**:
 - Terminal output
 - 7-segment display output

3. Architecture

- Must use **diamond inheritance**

Class Responsibilities

1. Stream (Base Class)

- Acts as a **common interface**.
- May contain shared functionality or virtual destructor.
- Must be inherited **virtually** to avoid diamond duplication.

2. stream

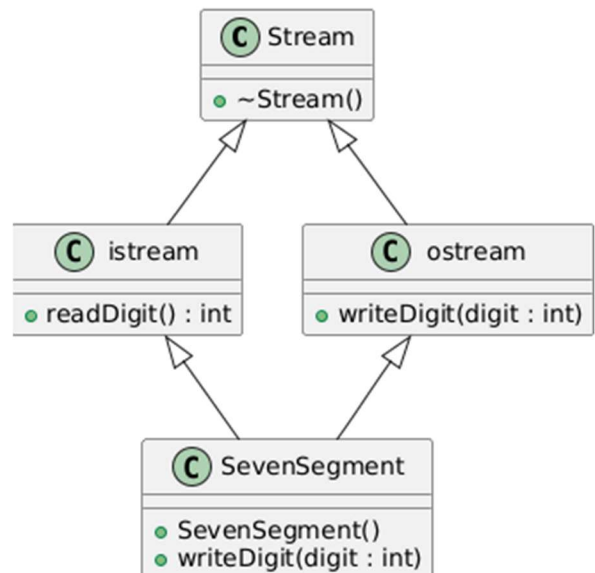
- Responsible for **reading input from terminal**.
- Ensures the input is a valid digit (0–9).
- Does **not** know anything about output.

3. ostream

- Responsible for **output abstraction**.
- Declares a generic output method (e.g., writeDigit(int)).
- Does **not** decide where the output goes.

4. SevenSegment

- Inherits from **both istream and ostream**.
- Implements output specifically for a **7-segment display**.
- May use GPIO or mock logic depending on environment. **(ONLY FOR STUDENT THAT DOESN'T HAVE HW)**



Compilation Process

Below is a **complete, generic project structure, class file split (include/src), library-based build**, and a **CMake setup** where:

- Each class has its own **header + source**
- All classes are compiled into a **shared library, (add_library) & (add_subdirectory)**
- main.cpp is built as an **executable (add_executable)**
- The design remains **generic** (student implements logic)

```
SevenSegmentProject/  
|  
├─ CMakeLists.txt  
|  
├─ include/  
|   ├─ Stream.hpp  
|   ├─ IStream.hpp  
|   ├─ OStream.hpp  
|   └─ SevenSegment.hpp  
|  
├─ src/  
|   ├─ Stream.cpp  
|   ├─ IStream.cpp  
|   ├─ OStream.cpp  
|   └─ SevenSegment.cpp  
|  
└─ app/  
    └─ main.cpp
```

Make sure to make the target_include_directory provided as needed to the user application.